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Back to primitives is the right way into the future

If you are game for more introspection, take a look at Debaleena Chattopadhyay's research paper (www.pages.iu.edu/~debchatt/wader/Chattopadhyay-Debaleena-ITS-2015-DC.pdf), which suggests exploration of interaction primitives to develop truly-intuitive gesture based interfaces.

Interaction primitives are the basic units that constitute an interface control. She writes, "Instead of assuming that our familiarity with everyday gestures in the physical world directly translates into our ability to perform those exact gestures in touch-less interfaces, we need to examine what makes a touch-less primitive motor intuitive. Or, which primitives allow unconscious application of our pre-existing sensorimotor knowledge (for example, visual perception or motor skills) during touch-less interactions?" This would result in more natural and accurate touch-less interfaces in the future.

from a ship or from the shore. When Jacobs Robotics Group's gesture-recognition system was mounted on the underwater robot and integrated into its control system, divers could directly control the robot underwater using sign language.

Gestures are vaguer than 0s and 1s

As useful as gestures are, these are not easy to implement in an HMI. In an ideal system that does not require wearables or controlled environments, and also does away with the support of touchscreens and other devices, the possibility of achieving unambiguous inputs is very bleak. Developing a simple system like that could involve immense efforts to not just develop the hardware but also to define an ideal set of gestures suitable to the user group. Because when a dictionary of accepted gestures is not defined, communication could be too vague and only another human would be capable of understanding it.

One example of such efforts comes from a group at IIT Guwahati, Assam, which aimed to standardise seven gestures to represent computational commands like Select, Pause, Resume, Help, Activate Menu, Next and Previous. The gestures were to be used by pregnant women watching a televised maternal health information programme in rural Assam. Participants belonged to the low socio-economic strata and most had

poor literacy levels.

A participative study deduced 49 different gestures that participants performed to represent the seven computational functions. According to the team, they selected seven body gestures based on frequency of use, logical suitability to represent functions, decreased possibility that the gestures would be accidentally performed (false positives) and ease of detection for the chosen technology (technical limitations).

If such a simple application requires such long research to arrive at the right gestures, you can imagine how much more complex the situation gets when a large and varied user base is expected, special audiences like disabled people or young children are involved, or when difficult environments like mines or oceans are involved.

Indeed, gesture based computing is a difficult dream to achieve, but it could take computing to several people who really need it. In a book totally unrelated to computers, American author Libba Bray wrote an oft-quoted line, "That is how change happens. One gesture. One person. One moment at a time."

This holds true for intuitive gesture interfaces, too. One gesture could save a man from drowning. One gesture could help a paralysed person live independently. One gesture could help a child learn about planets. One gesture could forge a business deal. One gesture could change the world! **EFY**